

Research Notes

1 Introduction

Document will outline my approach to learning hydraulics and how to create a hydraulic launcher. Hydraulic launchers are interesting because they are used often in rollercoaster launch systems, see figure 1, military drones, and aircrafts. Often, large drones are catapulted with hydraulic launchers because of their ability to deliver high force, and precise speed in a compact area.



Figure 1: *Wicked* rollercoaster from theme park in Utah [1].

I'll detail in this report the exact mechanics of hydraulics and how one can fabricate one from scratch.

2 Hydraulics

Hydraulics move heavy machinery, like construction equipment, around using trapped liquid. That might be ambiguous, but that trapped liquid compresses and exerts massive forces due to pressure. A good example is taking an empty syringe, placing one's finger on the end, and then pushing the plunger until it is physically unable to. When applied force on the plunger is released, the plunger springs back to its original position due to pressure. However, if that syringe is filled with water, the whole scenario becomes widely different, leading us into *Pascal's Law*, the direct mathematical foundation for hydraulics. But, first. let's analyze the physical properties of pressure and its relationship to energy.

2.1 Pressure

Pressure is defined as force per unit area, or:

$$P = \frac{F}{A} \quad (1)$$

and has the SI units Pascal, see figure 2, which are Newtons per square meter. It is often more convenient to use pressure rather than force to describe fluid behavior as they are more complicated systems.

For any object sitting on Earth, it experiences the force of gravity, and thus, the normal force exerted by the surface that it is resting on. That force of gravity is an object's weight, and the normal force is the feeling of having weight. However, for non-point masses, different orientations may yield different areas in contact with the surface, which yields different exerted pressures. Consider a 20 kg box with length 5 metres, width 10 meters, and height 2 meters. If you lay the box on its large side, the area in contact is 50 m^2 , and has a pressure computed out to:

$$P = \frac{200 \text{ N}}{50 \text{ m}^2} = 4 \text{ Pa}$$

But, if the box is laid on a more narrow side, the area in contact is 20 m^2 and the pressure then becomes 10 Pa . Intuitively, it makes sense why the pressure would be more. Standing on our feet, the pressure is much more manual, but if you try to balance on one finger, even just for a second, the pain on that hand becomes unbearable. When the area of contact decreases, the pressure increases. This explains why sharper objects cut easier with less amounts of force. When the tip of a blade is sharpened so that the area of contact is as small as possible, it cuts through materials like butter.

For non-solid objects, pressure is considered differently. For liquids, the medium is treated as a continuous distribution of matter, whereas a gas' pressure is considered to be the average pressure from molecular collisions with the wall. Think *Ideal Gas Law*: $PV = nRT$. When the number of moles increases, there are more gas molecules hitting the wall, increasing the pressure

Due to the nature of equation (1), adding another distance multiplier can change the nature of the equation. Let's consider the box example again. Remember that work is defined as the force applied parallel to the displacement of an object, or:

$$W = F \cdot d \quad (2)$$

and that if adding an additional displacement modifier to area makes the unit m^3 , indicative of volume, where:

$$V = A \cdot d$$

then, one can substitute both of these equations into the numerator and denominator respectively for equation (1), so that:

$$P = \frac{F \cdot d}{A \cdot d} = \frac{W}{V} \quad (3)$$

where the displacement vector cancels. Then, by the definition of the third term in equation (3), pressure can be defined as a measure of the energy per unit volume of an object. Since energy is *stored* per unit volume when pressure is applied, that alludes to the fact that there is a kinetic and potential energy associated with fluids.

If pressure is able to be defined by the energy per unit volume of a liquid, then pressure should be able to be defined as kinetic energy per unit volume of a liquid, as well as potential energy per unit volume of a liquid. If a liquid is moving, then the standard kinetic energy equation for moving objects, being:

$$KE = \frac{1}{2}mv^2 \quad (4)$$

can be substituted into the W portion of equation (3), so that:

$$P = \frac{\frac{1}{2}mv^2}{V}$$

Remember that density is described by how much mass fits into some specific volume, or mathematically:

$$\rho = \frac{m}{V} \quad (5)$$

which can then be substituted in the prior equation where:

$$P = \frac{1}{2}\rho v^2 \quad (6)$$

Furthermore, one can define pressure in terms of potential energy per unit volume as well, which can be used by deriving the potential energy for a mass suspended midair. If work equals the force acting parallel to the displacement to an object, as seen in equation (2), and if intuitively, the object does fall some amount of height h , then the potential energy stored in that object is:

$$U = mg \cdot h \quad (7)$$

which can then be substituted into equation (3). The reason this is the case is because, technically, water molecules are subject to gravity and will tend to *fall*, thus, storing energy. With further consideration to the density equation (5), pressure is defined as:

$$P = \rho gh \quad (8)$$

This equation is also known as the hydrostatic pressure of a liquid, which is the pressure exerted by a fluid at rest or in equilibrium, caused by the force of gravity. It is the weight of a liquid pushing down on itself and all other walls that surround it, and explain why it is unsafe to go deep into the ocean because hydrostatic pressure becomes so high. It also explains why one's ears will pop if they dive into water. When going deeper into water, the hydrostatic pressure increases overhead, causing a subsequent equalizing reaction of the ears to balance pressure.

2.2 Pascal's Law



Figure 2: French mathematician, philosopher, physicist, and prose writer; discovered Pascal's Law [2].

Considering the syringe scenario again, when the syringe is instead filled with a liquid such as water, the fluid is not as compressible in comparison to air, and begins to exert its pressure forces on the exterior of the syringe. Pascal's law restates this phenomena, claiming that any pressure applied to a confined liquid has its value project equally and undiminished (does not weaken) in all directions to every part of the whole fluid and the walls of its container.



Figure 3: Pressure is applied due to the stopper pushing down on the bottle [3].

Imagine a champagne bottle, like one seen in figure 3, has a stopper applying a constant force to keep the champagne contained. Let's say that the force the stopper is exerting on the tip of the bottle is equal to 10 N , and the area is 5 cm^2 . Therefore, with help from equation (1), the pressure at the tip is equal to $20,000\text{ Pa}$ or 20 kPa . With attention to Pascal's Principle, the pressure should then be exerted to every part of the liquid as well as the system that encloses it. Then, the sides of the bottle also experience a pressure equal to 20 kPa . What, though, is the force exerted on the sides of the bottle.

Let's consider the bottom of the bottle to have a pressure of 20 kPa with a surface area equal to 50 cm^2 . Rearranging equation (1) for force gives us:

$$F = PA \quad (9)$$

where the force at the bottom of the bottle is equal to 100 N . Wait, the stopper of the bottle was exerting a force of 10 N , how is it possible that there is more force at the bottom of the bottle than at the top. Remember that pressure goes undiminished, not force, meaning that there can be different forces at different parts of the bottle with the same pressure. Pascal actualized this through an experiment with a long narrow tube attached to a barrel. When the tube's height increased, the hydrostatic pressure increased as well, which was consistent with equation (8). However, as height increased, the barrel became more subject to bursting, until it did. Hence, Pascal realized that pressure was increasing throughout the entire system, but the force that was applied to the walls of the barrel changed.

Knowing that force becomes multiplied when pressure is the same for a container, but different for specific areas of the container, is the driving *force* for the concept of hydraulics. Moreover, it should be noted that at the bottom of any fluid container, there is some hydrostatic pressure overhead due to all the water molecules affected by gravitational potential energy. The pressure at the bottom of the container should then become:

$$P_{bottom} = P_{top} + \rho gh \quad (10)$$

2.3 Hydraulic Press

Now, we'll analyze a common hydraulic press scenario to illustrate the mathematics behind them. Consider an apparatus such as one seen in figure 4, where there is a hydraulic press attempting to move a car. m_{car} , d_1 , and d_2 are given, and we are asked to evaluate what force F_{piston} is required to move the car a distance d_2 .

Due to the conservation of energy principle, we can assert that the energy input of the system is equal to the energy output of the system, with the neglecting of non-conservative forces, so that:

$$W_{in} = W_{out} \quad (11)$$

And, with the equation (2), this equation expands out to:

$$F_{piston} \cdot d_1 = m_{car}g \cdot d_2 \quad (12)$$

where the right-hand side is gained by the acknowledgement that the weight of the car is being moved by a distance d_2 . The dot products of the forces and distances can be replaced by standard multiplication since the force is acting directly parallel to the displacement, and moreover, the equation (12) can be manipulated to solve for F_{d_2} where:

$$F_{piston} = m_{car}g \frac{d_2}{d_1} \quad (13)$$



Figure 4: Hydraulic press lifting a car (excuse the crude drawing).

This introduces a very important fact about hydraulics, in that, if you want F_{piston} to be small (in order to export a large amount of force), you must make it so that d_1 is a large value. If we introduce some numbers and say that the car is a *2025 Toyota Camry* with a mass approximately 1600 kg , then, in order to move the car up 5 m , with d_1 of 30 m , the force applied to the first piston should equal 2613 N . Although this force is noticeably smaller than the weight of the car, it is still around the weight of a grizzly bear. However, if we make it so d_1 is instead 100 m , then the force applied to the first piston becomes equal to 784 N .

This is still not what we want! If we want hydraulics to be powerful, we should derive another way to decrease the force applied to make the *2025 Toyota Camry* to move upwards by 5 m . With that, we'll claim that r_1 and r_2 radii of the pistons respectively are given, and that the pistons have the surface area conforming to the area of a circle. Due to the equation (1), there is a pressure exerted undiminished and equally to every part of the system by the car, where:

$$P_{car} = \frac{m_{car}g}{\pi r_2^2} \quad (14)$$

Then, there is some force being exerted on the surface area of piston #1 where:

$$F_{piston} = P_{car}\pi r_1^2$$

which after have its pressure term expanded due to equation (14), the force applied to the piston becomes:

$$F_{piston} = m_{car}g \frac{r_1^2}{r_2^2} \quad (15)$$

Let's lock in the F_{piston} in equation (15) to be 100 N , equivalent to lifting a large watermelon. Let's then revert the initial distances d_1 and d_2 to 30 m and 5 m respectively. Then, it seems intuitive to change r_1 and r_2 to get the intended 100 N I desire.

For incompressible fluid, any push in any direction must be followed by an equal and opposite push in a different direction. In other words, because there is no compressing, the mass of the fluid stays the same, and because the amount of fluid doesn't change, the volume of the fluid should not either. This concept is known as the conservation of volume, which holds that:

$$V_{in} = V_{out} \quad (16)$$

So, in this hydraulic press scenario, the volume changes that occur when a piston is pushing down are equal to the cross-sectional area times the displacement of the piston. If the first piston is pushed down, then the second piston is pushed up, which yields a corresponding volume change as well. Thus, using equation (16), we can see that both volume changes must equal each other to satisfy the conservation of volume, where:

$$\pi r_1^2 d_1 = \pi r_2^2 d_2 \quad (17)$$

which yields a non-linear equation where r_1 and r_2 must satisfy:

$$\frac{r_2^2}{r_1^2} = \frac{d_1}{d_2} \quad (18)$$

Now, that means there is some constraint to the situation, and that, we cannot get exactly 100 N pushing down on the piston. We will always get a F_{piston} that is consistent with a computation from the equation (13), and that the smallest possible force I can get is constrained by the distances the pistons travel by, which is actually a good conclusion to changing design constraints.

2.4 Hydraulic Brakes

2.5 Hydraulic Jacks

References

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